

Volatility is rough

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PUBLICATION YEAR	2018 (first circulated as arXiv:1410.3394, October 2014)
JOURNAL / WORKING PAPER	Quantitative Finance, 18(6), 933–949
DOI	10.1080/14697688.2017.1393551
RESEARCH AREA	Volatility modelling; high-frequency financial econometrics; derivatives
SERIES	Time Series
PAPER TYPE	Mixed (empirical and model-building, with econometric and probabilistic content)
REVIEWED BY	Arthur Faugeron
DATE REVIEWED	3 October 2026

ANALYTICAL ABSTRACT

The paper asks how smooth the latent volatility process of an asset really is. Using one-hour integrated-variance estimates for DAX and Bund futures and daily realized variance for the S&P 500, NASDAQ and nineteen further indices, the authors show that the moments of log-volatility increments scale as a power of the lag with an exponent proportional to the moment order, which identifies a single smoothness parameter H between roughly 0.06 and 0.2, and that the increments are close to Gaussian. They therefore model log-volatility as a fractional Brownian motion with H below one half, embedded in a fractional Ornstein-Uhlenbeck process with a very slow mean reversion (the Rough Fractional Stochastic Volatility model). Simulations from this model reproduce the empirical log-log regressions and also reproduce the apparent long memory found by classical tests, so the authors argue that persistence is spurious. A one-parameter forecast of log-variance beats AR and HAR benchmarks, and a Hawkes-process argument links roughness to order splitting and high-frequency trading. The main critical question is whether the roughness is a property of volatility or of the noisy proxies used to measure it.

1 Introduction

Source note. The review is built on the complete text of the arXiv version of the paper (arXiv:1410.3394, read in full including both proofs, the appendices, Tables 1 to 5 and Figures 1 to 19). The journal version in Quantitative Finance (2018) was not available as a readable document in this session, so statements about the paper refer to the arXiv text; wording, numbering and some figures may differ in the published version. The numbers attributed to the paper below were checked against that arXiv text. Items labelled Replication or illustrative calculation were computed for this review and are not results of the paper.

1.1 What the paper is about

PAPER Gatheral, Jaisson and Rosenbaum (2018) revisit a basic modelling question in derivatives and risk management: how regular are the sample paths of the instantaneous volatility process σ_t ? Black and Scholes (1973) take volatility as constant, local volatility models take it as a deterministic function of price and time, and the classical stochastic volatility models of Hull and White, Heston and SABR take it to be a diffusion, so that its paths are as irregular as a Brownian motion (Hölder exponent just below 1/2). The long-memory literature that followed Ding, Granger and Engle (1993) and Andersen, Bollerslev, Diebold and Labys (2003) moved in the opposite direction, towards smoother and more persistent volatility, using fractional Brownian motion with Hurst exponent H above 1/2 (Comte and Renault 1998). The paper's central claim is that both families are wrong about smoothness: log-volatility behaves like a fractional Brownian motion with H of order 0.1, that is, it is rougher than a Brownian motion.

CRITIQUE The paper is an unusual mixture. Its core empirical object is not a regression with a causal coefficient but a scaling law, so the notion of identification is different from that in a corporate-finance or asset-pricing paper: the information comes from how a moment of the increments changes as the lag changes, and the main threat is not endogeneity but measurement of an unobservable. This framing guides the whole review.

1.2 Executive understanding

WHAT IS THE PAPER TRYING TO DISCOVER?

Whether the regularity of log-volatility is described by a smoothness exponent H smaller than 1/2, whether that exponent is stable across assets, moments and sub-samples, and whether a model with only this property reproduces the facts usually attributed to long memory.

WHY DOES THE QUESTION MATTER?

The smoothness exponent controls three quantities that practitioners care about: how fast volatility can change over a day or a week (hedging and risk), how far ahead past volatility is informative (forecasting and margin setting), and the shape of the implied volatility surface near expiry (option pricing). In the paper the link to option markets is made through a result of

Fukasawa (2011): a stochastic volatility model driven by fractional Brownian motion with exponent H produces an at-the-money skew that behaves like $\tau^{H-1/2}$ at short expiries τ . The observed skew decays approximately as $\tau^{-0.4}$ (their Figure 2), which would require H close to 0.1.

WHAT WAS KNOWN BEFORE, AND WHAT WAS MISSING?

Before the paper, the stylized facts were: log-volatility increments are roughly Gaussian (Andersen et al. 2001), volatility is persistent with slowly decaying autocorrelations (Ding, Granger and Engle 1993), and a fractional model with H above $1/2$ was therefore the natural description. What was missing was a direct look at the smoothness of the path using high-frequency volatility estimates, and an explanation of why long-memory tests keep finding long memory if volatility is in fact rough.

WHAT DOES THE PAPER PROPOSE AND FIND?

It proposes the Rough Fractional Stochastic Volatility (RFSV) model, a fractional Ornstein-Uhlenbeck log-volatility with H below $1/2$ and a mean-reversion speed so small that, on any sample of practical length, the process is indistinguishable from a fractional Brownian motion. It finds H between 0.06 and 0.2 across 23 indices and contracts, shows the model reproduces the empirical scaling plots and the spurious long-memory statistics, and obtains improved variance forecasts with a single parameter.

2 Research Problem and Hypothesis

2.1 Research question

INFERENCE Stated precisely, the paper asks whether the lag- Δ increments of log-volatility, measured from high-frequency data, satisfy $E|\log \sigma_\Delta - \log \sigma_0|^q = K_q \Delta^{qH}$ for a single H below $1/2$ that does not depend on q , and whether a stationary model built on this property accounts for (i) the autocorrelation structure of volatility, (ii) the apparent long memory detected by classical procedures, and (iii) the out-of-sample predictability of future variance. The hypothesis is therefore a joint one: monofractal scaling with H well below $1/2$, Gaussian increments, and negligible mean reversion on the sample horizon.

2.2 Motivation

A reader should care for three reasons. First, the exponent is a primitive for pricing: if the market skew follows $\tau^{-0.4}$ and a fractional model with H near 0.1 delivers $\tau^{H-1/2}$ from a small number of parameters, then the same parameters can describe the historical dynamics and the shape of the option surface, which the Heston and SABR families cannot do in a time-homogeneous way. Second, the paper challenges a stylized fact that is more than twenty years old and is embedded in volatility forecasting practice (HAR, FIGARCH, fractional models). Third, it offers a statistical-physics style statement, power-law scaling with a single exponent, for a quantity that economists usually describe with autoregressions.

2.3 Existing literature

The paper positions itself against three bodies of work. The first is the diffusion literature on stochastic volatility (Hull and White 1993, Heston 1993, Hagan et al. 2002) and local volatility (Dupire 1994). The second is the long-memory literature (Ding, Granger and Engle 1993; Andersen and Bollerslev 1997; Andersen et al. 2001, 2003; Comte and Renault 1998), in which the autocorrelation function of absolute returns or volatility decays hyperbolically. The third is a set of structural and microstructure results: Hawkes-process fits to order flow whose kernels are nearly critical with power-law tails (Hardiman, Bercot and Bouchaud 2013; Bacry and Muzy 2014; Filimonov and Sornette 2012), and the scaling limits of such processes (Jaisson and Rosenbaum 2015, 2016). Competing explanations the paper acknowledges include the multifractal random walk of Bacry and Muzy (2003) and the sceptical view of Mikosch and Stărică (2000) that apparent long memory in returns may be an artefact of non-stationarity.

2.4 Research gap

Two gaps are identified by the authors. The long-memory evidence predates the electronic high-frequency era and, as Cont (2007) is quoted as saying, the short-range versus long-range dependence debate had not been settled. Second, no model simultaneously matched the smoothness of the volatility path, the term structure of the ATM skew, and the apparent long memory. A third gap, which the authors do not state, is the question of how sensitive the smoothness estimates are to the way spot volatility is proxied; this is the subject of section 7.

2.5 Claimed contribution and assessment

The claimed contributions are: the measurement of H (section 2); the RFSV model and its two limit results (section 3); the spurious long-memory argument (section 4); the forecast (section 5); and the microstructure link (section 6). **CRITIQUE** The first, second and fourth are convincingly demonstrated, within the limits of the proxy discussed below. The third is demonstrated only as consistency: it shows that two long-memory statistics produce similar output on simulated rough data and on real data, but it does not show that the same statistics fail to discriminate when applied to a genuinely long-memory model. The fifth is explicitly a heuristic, and the paper itself describes it as ideas that may help to understand the irregularity.

3 Theoretical Framework and Mechanism

The paper contains no equilibrium model with agents and preferences. Its mechanism is a statistical model of the volatility path together with a microstructure argument. This section reconstructs the mathematics in the order in which it is used: the building block (fractional Brownian motion), the empirical moment condition, the model, the two limit results, the covariance consequences, the forecast formula, the smoothing bias, and the Hawkes mechanism.

3.1 Fractional Brownian motion and what H controls

PAPER Let W^H be a fractional Brownian motion (fBM) with Hurst exponent $H \in (0,1)$: a centred Gaussian process with stationary increments and covariance

$$E[W^H_u W^H_s] = \frac{1}{2} (u^{2H} + s^{2H} - |u - s|^{2H}), \quad u, s \geq 0 \quad (1)$$

Setting $u = s$ gives $\text{Var}[W^H_u] = u^{2H}$, and expanding $E[(W^H_{t+\Delta} - W^H_t)^2]$ using (1) gives the scaling relation stated in the paper (their equation 1.1):

$$E[|W^H_{t+\Delta} - W^H_t|^q] = K_q \Delta^{qH} \quad (2)$$

K_q The q-th absolute moment of a standard normal variable, because each increment is Gaussian with variance Δ^{2H} .

Δ The lag, measured in days in the empirical work.

H The Hurst exponent. The paths are Hölder continuous of every order below H, so a small H means a jagged path.

The role of H is two-fold, and the paper's argument turns on separating the two roles. Locally, H sets the roughness: the increment over a short lag scales as Δ^H , which for $H = 0.1$ shrinks very slowly (a lag of one tenth of a day has typical increment $0.1^{0.1} \approx 0.79$ of the one-day increment, against 0.32 for $H = 1/2$). Globally, H sets the memory of the increments. From (1), the covariance of a unit increment with the increment k periods later is

$$\text{Cov}[W^H_p, W^H_k - W^H_{k-1}] = \frac{1}{2} (k^{2H} - 2(k-1)^{2H} + (k-2)^{2H}) \sim H(2H-1) k^{2H-2} \quad (3)$$

The sign of the factor $(2H-1)$ decides everything. For $H > 1/2$ the increments are positively correlated and the sum over k diverges, which is long memory in the technical sense used by Comte and Renault (1998). For $H = 1/2$ they are uncorrelated. For $H < 1/2$ the covariances are negative and summable: partial sums telescope to $\text{Cov}[W^H_p, W^H_n] = \frac{1}{2}(1 + n^{2H} - (n-1)^{2H})$, which tends to $1/2$ as n grows because $n^{2H} - (n-1)^{2H} \approx 2H n^{2H-1} \rightarrow 0$. In fBM, then, roughness and long memory are tied together through the single parameter H; the paper's innovation is to break the tie by making the process stationary at a scale far longer than the sample.

3.2 The empirical moment condition and the estimator of H

PAPER For a grid of mesh Δ on $[0, T]$ and $N = \lfloor T/\Delta \rfloor$ define the empirical moment of log-volatility increments:

$$m(q, \Delta) = (1/N) \sum_{k=1}^N |\log \sigma_{k\Delta} - \log \sigma_{(k-1)\Delta}|^q \quad (4)$$

The main maintained assumption (their equation 2.1) is that, for some $s_q > 0$ and $b_q > 0$, $N^{s_q} m(q, \Delta) \rightarrow b_q$ as Δ tends to zero. Under stationarity of the increments, $m(q, \Delta)$ is the sample analogue of $E|\log \sigma_\Delta - \log \sigma_0|^q$. If log-volatility were an fBM scaled by v , equation (2) would give

$$\log m(q, \Delta) = \zeta_q \log \Delta + \log(K_q \nu^q), \quad \zeta_q = q H \quad (5)$$

so that a regression of $\log m(q, \Delta)$ on $\log \Delta$ delivers a slope ζ_q and an intercept that identifies ν . The linear dependence of ζ_q on q is the test of monofractality: a multifractal process (Bacry and Muzy 2003) has ζ_q concave in q . The paper reports that the plotted ζ_q are very slightly concave but argues that the same concavity appears in simulated fBM samples of equal length, so it is a finite-sample effect. The estimated exponent reported in the tables is ζ_q/q , regressed over lags $\Delta = 1, \dots, 30$ days.

3.3 The RFSV model

PAPER Matching the empirical increment law gives a non-stationary model (their 3.1 and 3.2):

$$\log \sigma_{t+\Delta} - \log \sigma_t = \nu (W_{t+\Delta}^H - W_t^H) \Leftrightarrow \sigma_t = \sigma \exp\{\nu W_t^H\} \quad (6)$$

This is not stationary, since $\text{Var}[\nu W_t^H] = \nu^2 t^{2H}$ grows without bound. To restore stationarity the authors use the fractional Ornstein-Uhlenbeck process (Cheridito, Kawaguchi and Maejima 2003), the stationary solution of

$$dX_t = \nu dW_t^H - \alpha (X_t - m) dt \Rightarrow X_t = \nu \int_{-\infty}^t e^{-\alpha(t-s)} dW_s^H + m, \quad \sigma_t = \exp\{X_t\} \quad (7)$$

- ν Volatility of log-volatility at unit lag (equals the scale of the one-day increment because $\Delta^{2H} = 1$ at $\Delta = 1$).
- α Speed of mean reversion; $1/\alpha$ is the time scale of stationarity. The paper takes $\alpha \ll 1/T$, where T is the sample length.
- m Long-run mean of log-volatility.
- X_t Log-volatility; the stochastic integral is a pathwise Riemann-Stieltjes integral, well defined because fBM has Hölder paths.

This is the Comte and Renault (1998) fractional stochastic volatility (FSV) model with two parameter regions swapped: the original takes $H > 1/2$ to generate long memory and a large α to produce a decaying skew, whereas the RFSV model takes $H < 1/2$ and a very small α . The paper's economics for this choice is the following chain. Fukasawa (2011) implies that for H above $1/2$ the ATM skew $\psi(\tau)$ would be increasing in expiry for small τ , contradicting the observed decreasing skew. Comte and Renault could repair this only by choosing a large α , which makes the skew inconsistent at very short expiries. With H below $1/2$ the short-expiry skew $\tau^{H-1/2}$ is decreasing from the start without needing a large α . The choice of fOU rather than another stationary process is, in the authors' words, the simplest way to accommodate local fBM behaviour with stationarity; they do not claim that it is the only one.

PROPOSITION 3.1 AND ITS PROOF

The claim that fOU with tiny α is locally an fBM is formalised as follows. Integrating (7) by parts gives $X_t^\alpha = \nu W_t^H - \nu \alpha \int_{-\infty}^t e^{-\alpha(t-s)} W_s^H ds + m$, hence

$$(X_t^\alpha - X_0^\alpha) - \nu W_t^H = - \int_0^t \nu \alpha e^{-\alpha(t-s)} W_s^H ds - \int_{-\infty}^0 \nu \alpha (e^{-\alpha(t-s)} - e^{\alpha s}) W_s^H ds \quad (8)$$

Taking the supremum over $t \in [0, T]$ and using the maximal inequality of Novikov and Valkeila (1999) for fBM, the expectation of the supremum is bounded by $c(\nu \alpha T T^H + \int_{-\infty}^0 \nu \alpha (T \alpha e^{\alpha s}) |s|^H ds)$, which tends to zero as α tends to zero. So $E[\sup_{t \leq T} |X_t^\alpha - X_0^\alpha - \nu W_t^H|] \rightarrow 0$. Economically: if the mean-reversion time scale $1/\alpha$ is much longer than the sample, the restoring force is invisible over the sample and the path is an fBM plus a constant. Corollary 3.1 follows from this by uniform integrability, because Gaussian moments are uniformly bounded: $E|X_{t+\Delta}^\alpha - X_t^\alpha|^q \rightarrow \nu^q K_q \Delta^{qH}$.

CRITIQUE The proposition is a statement about convergence as α tends to zero, not a rate. It guarantees nothing about how small α must be relative to $1/T$ for the approximation error to be negligible at a given sample length. The simulations use $\alpha = 5 \times 10^{-4}$ per day, so $1/\alpha = 2000$ days against a simulated sample of 2000 to 3500 days: the reversion time scale is comparable to the sample length, not much larger. The approximation is therefore tested numerically in the paper, but not guaranteed by the proposition at those parameter values.

3.4 Covariance structure: why volatility is not a power law

PAPER For the Gaussian process X , the identity $E[(X_{t+\Delta} - X_t)^2] = 2 \text{Var}[X_t] - 2 \text{Cov}[X_t, X_{t+\Delta}]$ and Corollary 3.1 with $q = 2$ (where $K_2 = 1$) give Corollary 3.2:

$$\text{Cov}[X_t, X_{t+\Delta}] = \text{Var}[X_t] - \frac{1}{2} \nu^2 \Delta^{2H} + o(1) \quad (9)$$

The covariance of log-volatility is linear in Δ^{2H} rather than a power of Δ ; the paper's Figure 10 verifies the linearity for the S&P with $H = 0.14$. The exact finite- α expressions, taken from Cheridito et al. (2003) as quoted by the authors, are

$$\text{Var}[X_t] = H(2H - 1) \nu^2 \Gamma(2H - 1) / \alpha^{2H} = \nu^2 \Gamma(2H + 1) / (2 \alpha^{2H}) \quad (10)$$

where the second form uses $\Gamma(2H+1) = 2H(2H-1)\Gamma(2H-1)$. Since $\sigma = e^X$ and X is Gaussian, the product moment of volatility is lognormal. With $\mu = E[X_t]$, $E[\sigma_t \sigma_{t+\Delta}] = \exp\{2\mu + \frac{1}{2} \text{Var}[X_t + X_{t+\Delta}]\} = \exp\{2\mu + \text{Var}[X] + \text{Cov}[X_t, X_{t+\Delta}]\}$; substituting (9) yields their equation (3.6), $E[\sigma_{t+\Delta} \sigma_t] \approx \exp\{2\mu + 2 \text{Var}[X]\} e^{-\nu^2 \Delta^{2H}/2}$. Subtracting $E[\sigma]^2 = e^{2\mu + \text{Var}[X]}$ gives the form used in section 4:

$$\text{Cov}[\sigma_t, \sigma_{t+\Delta}] \approx C e^{-B \Delta^{2H}} - D, \quad B = \nu^2/2, \quad C = e^{2\mu + 2\text{Var}[X]}, \quad D = e^{2\mu + \text{Var}[X]} \quad (11)$$

This is a stretched exponential with a negative offset, which is neither a power law nor an exponential. The argument is that a log-log plot of the autocovariance of σ against Δ is not a straight line (their Figure 12), so the long-memory definition in terms of a hyperbolic decay cannot be satisfied.

ILLUSTRATIVE CALCULATION: THE IMPLIED AUTOCORRELATION OF Σ

EXTENSION Using the paper's simulation parameters ($H = 0.14$, $\nu = 0.3$, $\alpha = 5 \times 10^{-4}$), equation (10) gives $\text{Var}[X] = 0.340$, a standard deviation of 0.58 in log-volatility. The lognormal correlation of σ is $(e^{\text{Cov}} - 1)/(e^{\text{Var}} - 1)$. With Cov from (9), the autocorrelation of σ is 0.85 at one day, 0.76 at five days, 0.64 at 25 days, 0.45 at 125 days and 0.34 at 250 days (own calculation from the paper's formulae; the

approximation (9) is used at lags where $\alpha\Delta$ is small). The local log-log slope of this autocorrelation between successive lags is about -0.06 , -0.11 , -0.22 and -0.39 : it keeps steepening, so there is no straight line, yet the decay over this range is slow enough to look like a power law with exponent well below one on a coarse plot. This is the mechanism by which a statistical procedure written for power-law decay can wrongly report long memory, and it is the core of the section 4 argument.

3.5 The forecast formula

PAPER For an fBM with $H < 1/2$, Theorem 4.2 of Nuzman and Poor (2000) gives the conditional expectation of the future value given the past path, and by Corollary 3.1 log-variance can be treated as $\log \sigma_t^2 \approx 2\nu W_t^H + C$ over the sample, so

$$E[\log \sigma_{t+\Delta}^2 | F_t] = (\cos(H\pi)/\pi) \Delta^{H+1/2} \int_{-\infty}^t \log \sigma_s^2 / [(t-s+\Delta)(t-s)^{H+1/2}] ds = \\ (\cos(H\pi)/\pi) \int_0^\infty \log \sigma_{t-\Delta u}^2 / [(u+1)u^{H+1/2}] du \quad (12)$$

The second form follows from the substitution $s = t - \Delta u$. The formula is a weighted average of past log-variances with kernel $w(u) = \cos(H\pi) / [\pi(u+1)u^{H+1/2}]$, where u is the lookback measured in units of the forecast horizon. The weights integrate to one, a fact the paper does not state: by the Beta-function identity $\int_0^\infty u^{a-1}/(1+u) du = \pi / \sin(\pi a)$ with $a = 1/2 - H$, we get $\int_0^\infty du / [(u+1)u^{H+1/2}] = \pi / \cos(H\pi)$, which cancels the prefactor. The forecast is therefore a proper weighted average of past log-variances. Two limits are informative. As H tends to $1/2$ the prefactor $\cos(H\pi)$ goes to zero while the integral concentrates near $u = 0$, and the forecast collapses to the last observation, as it must for a Brownian motion. As H tends to zero the kernel becomes $1/(\pi \sqrt{u(1+u)})$, the multifractal-random-walk predictor of Duchon, Robert and Vargas (2012) that the authors cite.

The variance forecast adds a convexity correction. Conditional on F_t , $W_{t+\Delta}^H$ is Gaussian with variance $c \Delta^{2H}$, where $c = \Gamma(3/2 - H) / [\Gamma(H + 1/2) \Gamma(2 - 2H)]$ (Nuzman and Poor 2000), so log-variance has conditional variance $4c\nu^2 \Delta^{2H}$ and the lognormal mean is

$$E[\sigma_{t+\Delta}^2 | F_t] = \exp\{E[\log \sigma_{t+\Delta}^2 | F_t] + 2c\nu^2 \Delta^{2H}\} \quad (13)$$

For $H = 0.14$, $c = 0.695$ (own calculation), so the conditional standard deviation of log-variance is 0.50 at a one-day horizon, 0.63 at five days and 0.76 at 20 days when $\nu = 0.3$. The slow growth with the horizon (a factor 1.5 between one day and 20 days, versus 4.5 for a Brownian motion) is the reason forecasts at 20 days remain informative, and it is also the reason the benchmark ratio P in the tables does not approach one quickly.

3.6 Smoothing bias from integrated-variance proxies (Appendix C)

PAPER Spot volatility is not observed. The S&P and NASDAQ proxies are daily realized variances, which average the variance over a window of length δ . For the tractable fractional Stein-Stein model $v_{t+\Delta} - v_t = \kappa (W_{t+\Delta}^H - W_t^H)$ with $v = \sigma^2$ and the proxy $A_t^\delta = (1/\delta) \int_0^\delta v_{t+u} du$, the second moment of proxy increments is

$$m_{\delta}(2, \Delta) = (\kappa^2 / \delta^2) \int_0^{\delta} \int_0^{\delta} \{ |u - s + \Delta|^{2H} - |u - s|^{2H} \} du ds \quad (14)$$

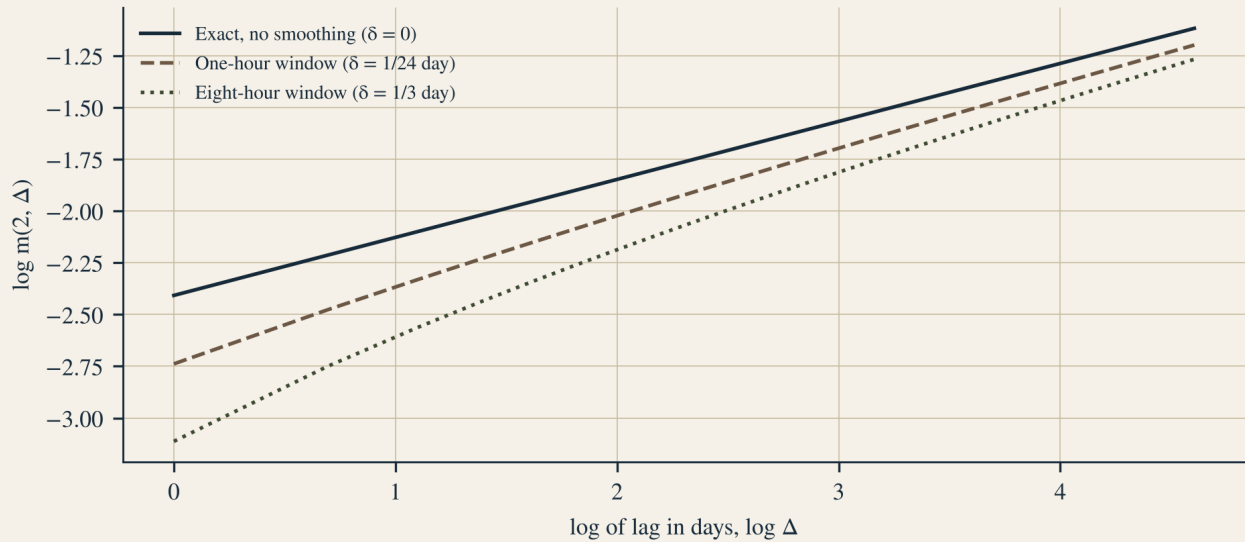
obtained by expanding the square, using (1) and symmetry. For $\Delta > \delta$ the integrals are elementary, and with $\theta = \delta/\Delta$ they give

$$m_{\delta}(2, \Delta) = \kappa^2 \Delta^{2H} f(\theta), \quad f(\theta) = [1 / ((2H+1)(2H+2))] \cdot [(1+\theta)^{2H+2} - 2 - 2\theta^{2H+2} + (1-\theta)^{2H+2}] / \theta^2 \quad (15)$$

The function f is below one and increases to one as θ falls to zero. Because $\theta = \delta/\Delta$ shrinks as Δ grows, the log-log regression of m_{δ} on Δ has an effective slope above $2H$, so H is biased upwards, and the intercept implies a lower effective v . This is the direction of bias in the paper's claim that whole-day proxies overstate H relative to one-hour proxies, and it means that the headline estimates of H from daily realized variance are upper bounds under this smoothing mechanism, if measurement noise is ignored.

REPLICATION Equation (15) was evaluated for $H = 0.14$, $\kappa = 0.3$ and $\Delta = 1, \dots, 100$, and $\log m_{\delta}$ was regressed on $\log \Delta$. The regression gives $(\kappa, H) = (0.300, 0.140)$ for no smoothing, $(0.263, 0.161)$ for a one-hour window ($\delta = 1/24$ day), and $(0.230, 0.184)$ for an eight-hour window ($\delta = 1/3$ day). These reproduce the paper's Table 5 exactly $(0.300/0.140, 0.263/0.161, 0.230/0.184)$, which confirms the appendix algebra.

Figure 1. Smoothing bias in the log-log scaling regression: m_{δ} vs Δ in the fractional Stein-Stein model with $H = 0.14$



Source: own calculation from equation (15) of this review, which is the paper's Appendix C formula; $H = 0.14$, κ lags Δ to 100 days. Generated, not taken from the paper. Higher smoothing windows raise the fitted slope (0.161 and 0.184 against the true 0.140) and lower the intercept.

3.7 The microstructure mechanism: Hawkes processes

PAPER Section 6 proposes an explanation for roughness. Let N_t count transactions and let its intensity be a Hawkes process

$$\lambda_t = \mu + \sum_{0 < J_i < t} \phi(t - J_i) \quad (16)$$

μ	Baseline (exogenous) arrival intensity.
J_i	Past event times.
ϕ	Non-negative kernel; its L^1 norm is the average number of events directly triggered by one event (the endogeneity ratio).

The authors cite two calibration regularities: the L^1 norm of ϕ is close to one (nearly critical, a highly endogenous market) and ϕ has a power-law tail (an order influences the flow for a long time, attributed to the splitting of large orders into metaorders). They state, citing Jaisson and Rosenbaum (2015, 2016), that the macroscopic scaling limit of such a process is an integrated fractional process with Hurst exponent below 1/2, and that integrated variance is linearly related to cumulated order flow. The conclusion is that volatility should behave as a fractional process with H below 1/2. **CRITIQUE** This is a heuristic chain of three assertions (near-criticality, heavy tails, linearity between order flow and variance), none of which is derived in this paper. It is not a test of the mechanism. The scaling-limit results themselves are proved in the cited papers and are where the real mathematics lies.

4 Data and Empirical Strategy

4.1 Where the identifying variation comes from

INFERENCE The paper has no treatment, instrument or exogenous shock. Its empirical content is a within-sample scaling relation: the same series is examined at many lags Δ , and the information about H comes from how a moment of the increments of log-volatility grows with Δ . Identification of H therefore rests on three assumptions: that the increments of log-volatility are stationary, that the proxy used for spot volatility differs from the truth only by a smoothing effect that can be simulated, and that the relationship is log-linear in the lag range used. The causal-identification questions of an asset-pricing paper do not arise, but their equivalent does: the estimate of H is a property of the proxy series, and equals a property of latent volatility only if the proxy error is benign.

4.2 Data

Table 1. Datasets used in the paper

Series	Proxy for spot variance	Sample	Days
DAX futures; Bund futures	Integrated variance, 10 am to 11 am London time, from the model with uncertainty zones (Robert and Rosenbaum 2011, 2012)	13 May 2010 to 1 August 2014	1,248
S&P 500, NASDAQ and 19 other indices	Daily realized variance from the Oxford-Man Institute Realized Library (5-minute returns, whole trading day)	3 January 2000 to 31 March 2014	3,540
Forecast sample	Five indices (S&P, FTSE, Nikkei 225, DAX, CAC 40), AR coefficients re-estimated on a rolling 500-day window	As above, from observation 500	about 3,040

Source: the paper, section 2 and section 5. The last row's day count is the review's arithmetic (3,540 less the 500-day estimation window), not a number stated by the authors.

The economic content of the proxies matters. The uncertainty-zones estimator exploits the discrete-price structure of very liquid futures to estimate the variance over a short one-hour window, which the authors treat as a good approximation of the instantaneous variance because one hour is small relative to the day-and-longer lags that are examined. The Oxford-Man series integrates the variance over the whole trading day, so it cannot resolve the intraday path and, as shown in section 3.6, regularises it. No transformation, filtering, winsorisation or handling of missing days is described in the text read; the one-hour window fixes the time of day at which volatility is sampled, which removes the intraday seasonal pattern but also discards most of the trading day.

CRITIQUE Four properties of the data limit what can be concluded. The futures sample covers only about four years, all after the global financial crisis and during the euro-area crisis. The index sample runs from 2000 and includes the 2008 crisis, in which the paper itself reports H to be higher. The two datasets use different proxies, so the comparison of 0.08 to 0.125 (futures) against 0.14 (indices) mixes asset and proxy effects, which the authors resolve only by simulation. Finally, all estimates are on a single, partly overlapping set of developed-market equity indices and two European futures: no single stocks, currencies, commodities or emerging-market data (other than a few indices such as the Bovespa, Mexican and Korean indices), and no tick-level validation of the realized variance measure.

4.3 Estimation and forecasting procedures

PAPER For smoothness, $\log m(q, \Delta)$ is regressed on $\log \Delta$ for $q = 0.5, 1, 1.5, 2, 3$ and lags $\Delta = 1$ to 30 days for the cross-index table, and the average over starting points is used for each Δ . H is read as the slope divided by q . For the sub-sample test the Oxford-Man data are split into two halves. For forecasting, the RFSV predictor (12) with a Riemann-sum approximation is compared at horizons 1, 5 and 20 days with the $AR(p)$ predictors ($p = 5, 10$, estimated in R on a rolling window of 500 days) and the HAR predictor of Corsi (2009):

$$F_t(\Delta) = K_0^\Delta + C_0^\Delta \log \sigma_t^2 + C_5^\Delta (1/5) \sum_{i=0}^5 \log \sigma_{t-i}^2 + C_{20}^\Delta (1/20) \sum_{i=0}^{20} \log \sigma_{t-i}^2 \quad (17)$$

and accuracy is measured by the ratio of the mean squared forecast error to the sample variance of the target (their definition of P , with the notation below),

$$P = \sum_k (\gamma_{k+\Delta} - F_k(\Delta))^2 / \sum_k (\gamma_{k+\Delta} - \gamma_{avg})^2, \quad \gamma_k = \log \sigma_k^2 \quad (18)$$

$F_k(\Delta)$ The forecast of $\log \sigma_{k+\Delta}^2$ made at date k (equation 17 for HAR; the other predictors are analogous).

γ_{avg} The empirical mean of log-variance over the whole sample.

so that $1 - P$ is an out-of-sample R^2 relative to the full-sample mean. The ratio is computed on proxies of the target, and the paper states that it treats the volatilities as perfectly observed.

CRITIQUE Three features of the forecast design need caution. The paper describes the AR and HAR coefficients as estimated on rolling windows, but does not say whether H and v for the RFSV predictor are re-estimated in real time; the text of section 5.2 says that v^2 is obtained from the intercept of a regression of $\log m(2, \Delta)$ on $\log \Delta$, without specifying the window. If H and v are full-sample estimates, the RFSV forecast uses information that the AR and HAR competitors do not have at time t , a look-ahead advantage, even though a single parameter is a small advantage. The denominator in P uses the full-sample mean of the target, which is itself look-ahead information, but it applies equally to all predictors. And no formal test of equal predictive accuracy (Diebold and Mariano) is reported, so the differences in Tables 1 and 2 are not accompanied by standard errors.

5 Results

5.1 Smoothness estimates

WHAT WAS TESTED, WHAT WAS FOUND

EVIDENCE On the DAX and Bund futures the regression slopes ζ_q are close to a straight line in q with $H = 0.125$ for the DAX and 0.082 for the Bund. On the S&P and NASDAQ the text reports $H = 0.142$ and 0.139 . Across the 21 Oxford-Man indices (their Table 3, $\zeta_2/2$ column) H ranges from 0.075 to 0.158 , with a cross-index mean of 0.121 (own calculation from the table). The spread across q for a given index is small (average 0.007 , maximum 0.038 for the STOXX 50), which is the monofractality property. Two observations deserve attention. The text and the table do not agree for the S&P: the text gives 0.142 and the table 0.124 to 0.128 . The text does not say what lag range produced the figure in the text, so the likely explanation is a different fitting range, which is an inference. And the statement in the introduction that the smoothness parameter lies between 0.08 and 0.2 is not exactly the table range (0.071 to 0.158 across q and indices, own calculation).

Table 2. Estimated smoothness $H = \zeta_q/q$ for selected indices (Oxford-Man realized variance, lags 1 to 30 days)

Index	$q = 0.5$	$q = 1$	$q = 2$	$q = 3$	First half	Second half
S&P 500 (SPX)	0.128	0.126	0.124	0.124	0.115	0.158
FTSE 100	0.132	0.132	0.131	0.127	0.140	0.156
Nikkei 225	0.131	0.131	0.132	0.133	0.083	0.134
DAX	0.141	0.139	0.136	0.132	0.154	0.168
Australia (AORD)	0.072	0.073	0.075	0.077	0.059	0.114
Hang Seng	0.079	0.079	0.080	0.082	0.072	0.129
Mexico (MXX)	0.077	0.077	0.075	0.071	0.068	0.118
Swiss (SSMI)	0.149	0.153	0.158	0.158	0.206	0.183
All 21 indices, mean	0.122	0.122	0.121	0.118	0.113	0.146

Source: the paper, Appendix B (Tables 3 and 4), values as printed; the last row is the review's arithmetic mean over the 21 rows of those tables. The half-sample columns use the realized-kernel series (rk) in the paper's Table 4, which is a different realized measure from the rv series in the first four columns.

EVIDENCE What the table shows. For q between 0.5 and 3 the estimated H is nearly constant within a row, so the increments of log-volatility scale in the same way across moments, which is the property that distinguishes a monofractal (fBM-type) process from a multifractal one. Across indices H varies by a factor of two (0.072 for Australia to 0.158 for Switzerland). Between the two sub-samples H is higher in the second half for 18 of 21 indices, with a cross-index mean of 0.113 in the first half and 0.146 in the second (own calculation), so H is not a constant of nature in this evidence. The authors attribute the second half's higher estimates to the financial crisis, but they offer no test and no explanation of mechanism.

ECONOMIC MAGNITUDE

EXTENSION The magnitudes translate into intuitive quantities. With $\nu = 0.3$ (the value used in the paper's simulations) and $H = 0.14$, the typical change in log-volatility is 0.30 over one day, 0.38 over five days, 0.47 over 25 days, 0.59 over 125 days and 0.81 over 1,250 days (about five years). For $H = 1/2$ with the same ν the corresponding numbers are 0.30, 0.67, 1.50, 3.35 and 10.6. The paper states the ratio $1250^{0.14} = 2.7$, which is correct. A five-year change in log-volatility of only 0.81 is the formal meaning of the claim that the volatility process over one day resembles the process over a decade: the coefficient characterising the law of the path rescaled to the unit interval is multiplied by only 2.7 across a factor of 1,250 in time.

5.2 Gaussian increments and the stationarity point

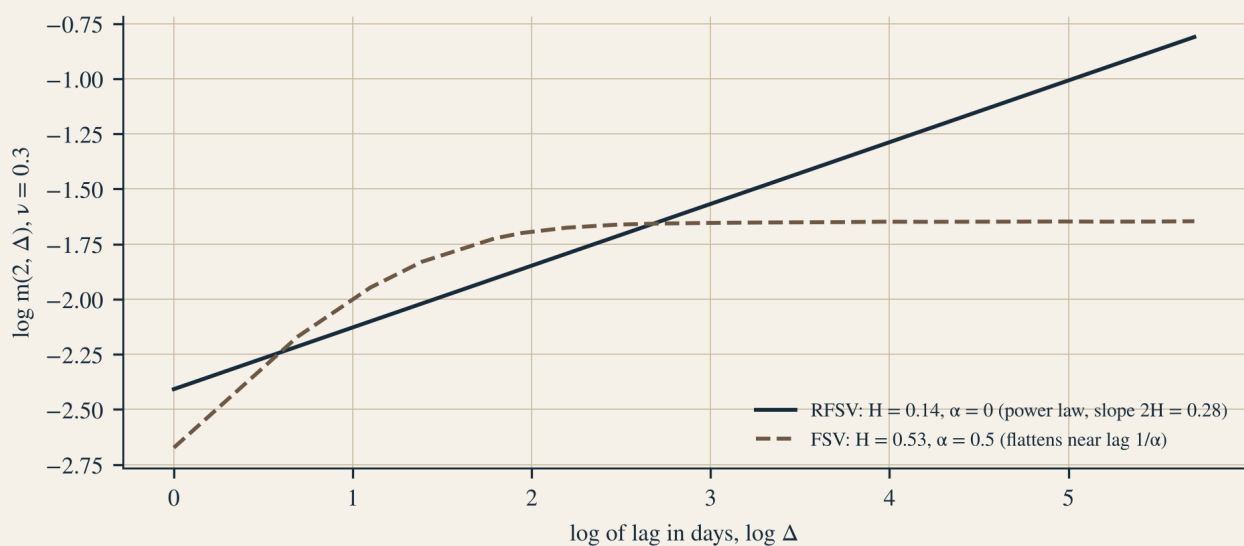
EVIDENCE Histograms of log-volatility increments at lags of 1, 5, 25 and 125 days (their Figure 9) are close to a fitted normal density, and the one-day normal fit rescaled by Δ^H matches the others. The authors note slight deviations and say they are consistent with those found for fBM samples of

similar length. The Gaussianity is visual; no formal normality test or quantile comparison is shown, which matters because the lognormal convexity correction in (13) and the covariance results (9) to (11) depend on Gaussianity in their entirety.

5.3 RFSV against FSV, and the skew

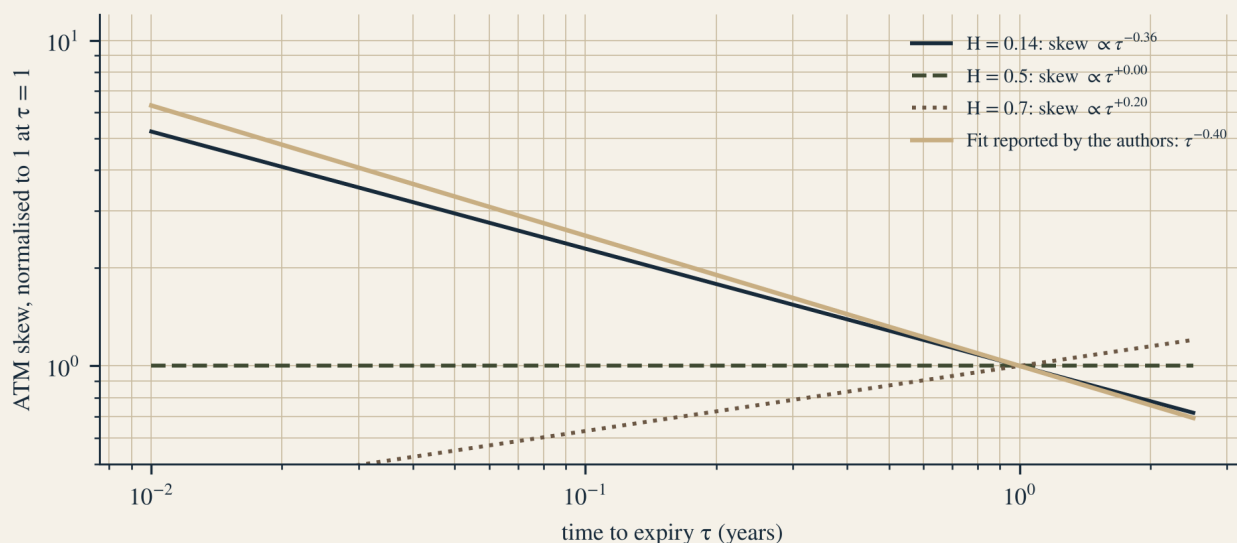
PAPER In Figure 13 the authors plot $m(2, \Delta)$ from the data against two models: the Comte-Renault FSV with $H = 0.53$ (the estimate of Chronopoulou and Viens 2012) and $\alpha = 0.5$, and RFSV with $\alpha = 0$ and $H = 0.14$. The conclusion is that long-memory FSV models cannot reproduce the lag structure: $m(2, \Delta)$ in FSV flattens at a lag of order $1/\alpha$, whereas in the data it continues to rise as a power of Δ with slope 2×0.14 .

Figure 2. Second moment of log-volatility increments, RFSV and FSV (illustrative parameters)



Source: own calculation. FSV: spectral representation of the fOU covariance (Cheridito, Kawaguchi and Maejima 2003), $H = 0.53$, $\alpha = 0.5$. RFSV: $\nu^2 \Delta^{2H}$ with $H = 0.14$, $\nu = 0.3$. Generated for this review; lags 1 to 300 days. The parameters of the FSV curve are those quoted in the paper's Figure 13; the value of ν is an assumption of this review and only shifts the curves vertically.

EVIDENCE The figure reproduces the logic of the paper's Figure 13. For the FSV curve, $m(2, \Delta)$ is 0.069 at one day, 0.187 at nine days, and 0.192 at 41 and 300 days (own calculation): it has saturated at twice the stationary variance, $2\text{Var}[X] = 0.193$, by the end of the first fortnight. The RFSV curve is 0.090 at one day, 0.167 at nine days, 0.255 at 41 days and 0.444 at 300 days. A least-squares line through the first six lags of the FSV curve has slope 0.25 (that is, an apparent H of 0.25, although the true H is 0.53), because mean reversion bends the curve even over lags of a few days. A researcher fitting an FSV model with H above $1/2$ to the short-lag data would therefore be pushed to a value of H far from the model value. This is consistent with the paper's claim that H must be set by the initial slope. It does not show that no long-memory model can match the data, only that this specific FSV parameter pair cannot.

Figure 3. Skew exponent implied by fractional volatility, against the observed decay

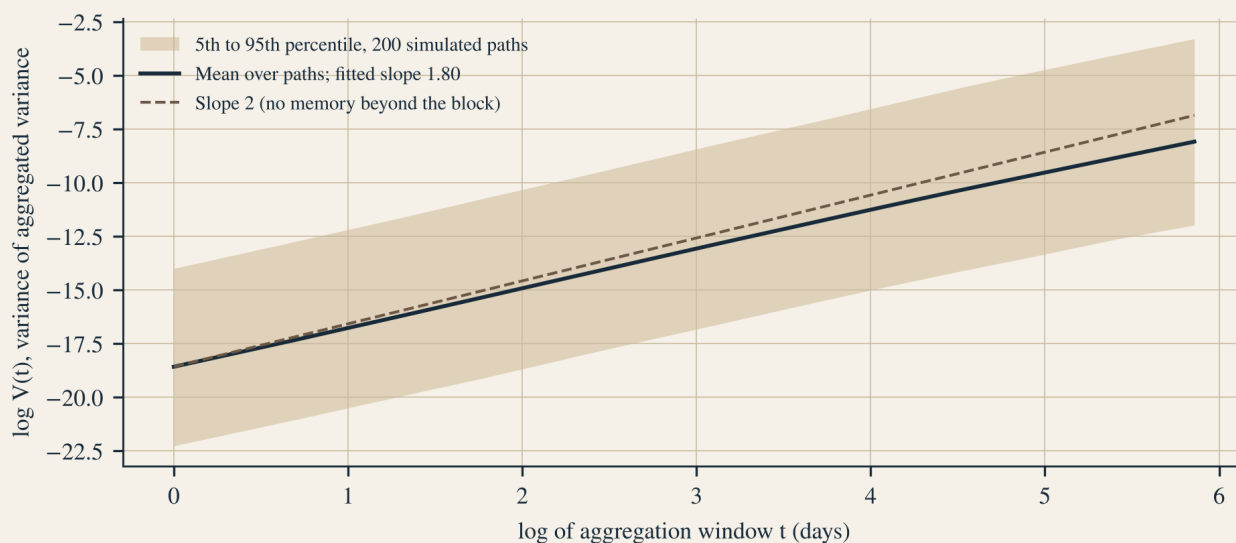
Source: own calculation. Curves show $\tau^{H-1/2}$ (Fukasawa 2011, as stated in the paper) for $H = 0.14, 0.5, 0.7$, normalised to one at $\tau = 1$ year; the thick line is the power-law fit $\tau^{-0.4}$ reported by the authors for the S&P 500 volatility surface of 20 June 2013 (their Figure 2). Generated, not data. The figure is a restatement of the asymptotic result, valid for small τ only.

EVIDENCE Figure 3 isolates the argument that links time-series roughness to the option surface. A skew proportional to $\tau^{H-1/2}$ is decreasing in τ only if H is below $1/2$; for the observed exponent of -0.4 , $H = 0.1$. $H = 0.14$ gives -0.36 , which is close to, but not exactly, the observed exponent. The gap between a time-series H of 0.12 to 0.14 and a skew-implied H of 0.1 is itself an empirical question that the paper leaves for future work.

5.4 Spurious long memory

PAPER The paper applies two long-memory procedures to the data and to simulations from RFSV ($H = 0.14$, $v = 0.3$, $\alpha = 5 \times 10^{-4}$, 3,500 days). The first is the Andersen et al. (2001) test: if $V(t) = \text{Var}[\int_0^t \sigma_s^2 ds]$ grows like $t^{2-\gamma}$ with γ below one, the autocorrelation of log-volatility decays like $t^{-\gamma}$. On both S&P data and simulated data the log-log plot is close to a line with slope 1.86 , implying $\gamma = 0.14$, which would be read as long memory. The second is the fractional-differencing check of Andersen et al. (2003): $\varepsilon_t = (1 - L)^d \log \sigma_t$ with $d = 0.4$ behaves as white noise on both data and simulation. The conclusion is that these procedures cannot distinguish genuine long memory from the rough, short-memory RFSV process.

REPLICATION The first test was repeated on 200 independent paths of 3,500 daily values of log-volatility generated as $X_t = -5 + 0.3 W^{0.14}_t$ (the $\alpha \rightarrow 0$ limit, by Cholesky decomposition of the exact fBM covariance matrix), with $\sigma^2 = e^{2X}$, $V(t)$ estimated as the variance of non-overlapping block sums for 14 window sizes from 1 to 350 days, and the slope from a log-log regression. The mean slope is 1.80 , with a 5th to 95th percentile range of 1.70 to 1.88 , which brackets the paper's 1.86 and implies, in the framework of Andersen et al., γ of about 0.2 . For comparison, a short-memory AR(1) log-volatility with a 20-day decorrelation time and similar overall variance gives a mean slope of 1.47 (range 1.37 to 1.56). The statistic therefore distinguishes strong short memory from RFSV, but its output for RFSV, a slope near 1.8 , is indeed what would be read as long memory.

Figure 4. Variance of aggregated variance in simulated RFSV paths

Source: own simulation (replication of the paper's Figure 16 procedure): 200 paths of 3,500 days, $H = 0.14$, $v \propto \alpha$ block sizes 1 to 350 days; random seed fixed. Generated, not data. The slope of a long-memory-free benchmark of 2 is shown dashed.

EVIDENCE The replication supports the paper's logic: a process with no hyperbolic decay yields an aggregated-variance slope well below 2. It does not show that the slope on actual data was produced by an RFSV process, only that it could be. The same caution applies to the paper's own use of this test.

5.5 Forecasting results

Table 3. Ratio P of mean squared error to variance of the target, averages over five indices

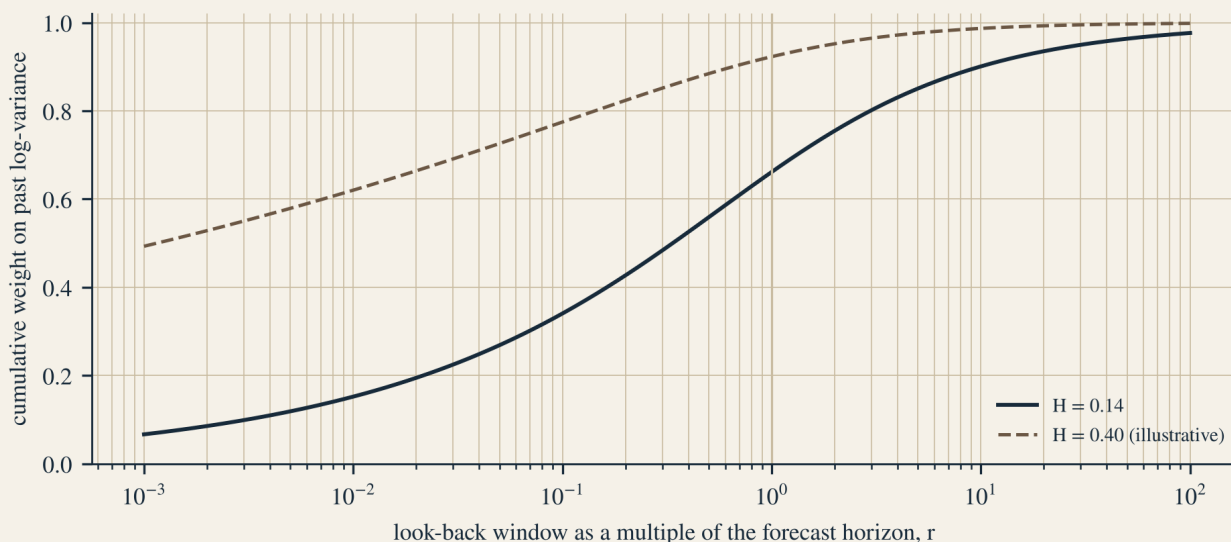
Target	Horizon	AR(5)	AR(10)	HAR	RFSV	RFSV against HAR
Log-variance	1 day	0.277	0.277	0.273	0.270	1.1% lower
Log-variance	5 days	0.424	0.412	0.401	0.388	3.2% lower
Log-variance	20 days	0.724	0.653	0.618	0.562	9.0% lower
Variance	1 day	0.532	0.551	0.489	0.480	1.9% lower
Variance	5 days	0.757	0.736	0.709	0.661	6.8% lower
Variance	20 days	1.025	0.968	0.942	0.863	8.4% lower

Source: the paper, Tables 1 and 2 (five indices: SPX, FTSE, N225, DAX, CAC), averaged by the review; the last column is $(1 - \text{RFSV}/\text{HAR})$ computed on the averages. In 29 of the 30 individual cells of the paper's two tables RFSV has the lowest P; the exception is the Nikkei 225 variance forecast at one day (RFSV 0.505, HAR 0.504).

EVIDENCE Because P is an inverse R^2 ($R^2 = 1 - P$), a log-variance forecast at 20 days explains on average about 44% of the variation of the target under RFSV, 38% under HAR and 28% under AR(5). The gain of RFSV over HAR grows with the horizon: about 1% at one day and 9% at 20 days. Forecast gains of this size are economically modest at short horizons but meaningful at the one-month horizon relevant for option and margin decisions. At 20 days the variance forecast has P

above 1 for AR(5) (1.025), meaning it performs worse than the unconditional mean of the target. The paper emphasises parsimony: one parameter H , against horizon-specific coefficient sets for AR and HAR. That is a real advantage, though the same parsimony argument would apply to a HAR model with a fixed coefficient pattern.

Figure 5. Weights of the RFSV forecast on past log-variance ($H = 0.14$ and $H = 0.40$)



Source: own calculation from the kernel in equation (12) of this review (their equation 5.1). Cumulative weight on observations from the last r forecast-horizons, with r on a logarithmic scale. Generated, not data. The curve for $H = 0.40$ is illustrative only.

EVIDENCE The kernel shows that for $H = 0.14$ about 66% of the weight lies within one forecast horizon of the present, 85% within five horizons and 90% within ten (own calculation). The authors say that for $r = 1$ the relevant tail integral is $\varepsilon = 0.35$; with the prefactor included the mass beyond one horizon is 0.34, in agreement, whereas the integral exactly as they define it (without the prefactor) equals 1.18. The statement is therefore correct for the normalised weights and imprecisely defined in the text. The practical implication, that a one-month forecast should mainly use the last month, is intuitive and matches the findings that motivated HAR.

5.6 Reading the main exhibits

Table 4. What the principal exhibits establish and fail to establish

Exhibit	Question and specification	Key reading	What it does not establish
Figures 3 to 8 of the paper	Does $\log m(q, \Delta)$ grow linearly in $\log \Delta$ Regression across lags for each q on DAX, Bund, S&P and NASDAQ	Straight lines; slope proportional to q ; H of 0.08 to 0.14	That the proxy error is negligible. Standard errors of H are not reported
Figure 9 of the paper	Is the increment distribution Gaussian at all lags?	Close to normal and consistent with Δ^H rescaling	Tail behaviour; no formal test

Exhibit	Question and specification	Key reading	What it does not establish
Figure 10 and Figure 11	Is Cov of log-volatility linear in Δ^{2H} ? Is $\log E[\sigma\sigma]$ linear in Δ^{2H} ?	Yes, over the range plotted for $H = 0.14$	That alternative functional forms (for example a two-factor exponential) cannot also be linear in Δ^{2H} over this range
Figure 13	Can FSV ($H = 0.53$, α) match $m(2, \Delta)$?	No: flattens at lag of order $1/\alpha$	That no long-memory specification does
Figures 16 and 17	Do classical long-memory procedures reject on simulated RFSV paths?	Same output on data and on RFSV simulation	That RFSV is the true generator; only that the procedures are not discriminating
Tables 1 and 2	Out-of-sample P ratio, RFSV against AR and HAR, 15 index-horizon pairs for each target	RFSV lowest in 29 of 30 cells; gain rises with horizon	Statistical significance of the differences; real-time parameter estimation
Tables 3 and 4 (Appendix)	H across 21 indices and across halves of the sample	0.07 to 0.16 across indices; higher in the second half	A mechanism for the shift in the second half
Table 5 (Appendix)	Effect of averaging window on the regression	H rises from 0.140 to 0.161 (1 hour) and 0.184 (8 hours)	Effect of microstructure noise on top of smoothing

Source: review, based on the paper's exhibits as read in the arXiv text.

6 Robustness and Additional Tests

The paper is not organised around a robustness section; its checks are embedded in the argument. They are collected below in the form concern, test, result, and whether the test resolves the concern.

Table 5. Robustness checks reconstructed from the paper

Concern	Test in the paper	Result	Does it resolve the concern?
Daily realized variance overstates smoothness (averaging)	Compare one-hour futures proxies with whole-day proxies; simulate RFSV and both estimators; Appendix C algebra	Simulated H of 0.16 (one hour) and 0.18 (eight hours) against the true 0.14; Table 5 reproduced here	Resolves the averaging bias in direction and size for the simulated design. It does not address measurement noise, which would bias H in the opposite direction
Proxy of spot volatility is itself an estimate	Simulation with discrete prices (uncertainty zones, tick 5×10^{-4} η 0.25)	Estimator recovers H within 0.02 to 0.04	Partly. The simulated microstructure is the same model that is used to build the proxy, so the test is partly self-referential

Concern	Test in the paper	Result	Does it resolve the concern?
H depends on moment order q (multifractality)	Compare ζ_q with qH for q from 0.5 to 3; compare with simulated fBM of equal length	ζ_q slightly concave in data and in simulations	Reasonably, for the range of q tested; it does not test q above 3
H is not stable in time	Split the sample into two halves (Table 4 of the paper)	H higher in the second half for 18 of 21 indices	Documents the concern but does not resolve it; a time-varying H weakens the single-parameter claim
Result specific to one asset	DAX, Bund, S&P, NASDAQ plus 19 other indices	Same scaling law everywhere, H from 0.07 to 0.16	Yes, across developed and some emerging equity indices; not across asset classes
Spurious long memory is an artefact of the RFSV simulation design	Two procedures run on both data and simulation (Figures 16, 17)	Same output	Resolves the narrow claim (the procedures cannot discriminate). Does not rule out a model that is both rough and persistent, as found later by Bennedsen, Lunde and Pakkanen (2022)
Forecast gain is spurious or overfit	Rolling 500-day window for AR; five indices; three horizons; log-variance and variance targets	RFSV lowest in 29 of 30 cells	Partly. No significance tests; unclear whether H, v are estimated in real time
Non-Gaussian increments or jumps	Visual histograms (Figure 9)	Close to Gaussian	No. Jumps in the price are not separated from continuous volatility variation in the realized-variance proxy

Source: review.

CRITIQUE The strongest tests are those that run the same statistical procedure on real and simulated data from the proposed model, because they check that the model can generate what the data show. The weakest point of the set is that none of them alters the information in the proxy: every test uses the same type of volatility estimate as input, so no check separates a property of latent volatility from a property of its estimator.

7 Critical Evaluation

7.1 What the paper proves

ESTABLISHED

High-frequency proxies of log-volatility for 23 series exhibit power-law scaling of the moments of their increments over lags of one to about thirty days, with an exponent approximately proportional to the moment order (monofractality) and with H between about 0.07 and 0.16. Within the model, the fOU log-volatility with H below $1/2$ and very small α has the second-moment and covariance structure of equations (9) to (11), which match the empirical linearity in Δ^{2H} . The classical long-memory procedures discussed (Andersen et al. 2001, 2003) return long-memory-like output on RFSV simulations, as the replication in section 5.4 confirms. The smoothing bias of integrated-variance proxies runs upwards, as the Appendix C algebra and its replication show.

SUPPORTED BUT CONDITIONAL

That H below $1/2$ describes latent volatility rather than the proxy: it holds only if measurement error in the proxies does not itself generate apparent roughness. That RFSV forecasts beat AR and HAR: it holds for MSE on log-variance and variance, five indices, if H and v are estimated without look-ahead. That variation of H across indices and sub-periods is small enough for a single-parameter forecast: the data show a shift from about 0.11 to 0.15 between halves.

SUGGESTIVE

That a rough volatility model fits the implied volatility surface (asserted for future work; only the short-expiry asymptotic $\tau^{H-1/2}$ is used). That order splitting and high-frequency trading cause roughness (a heuristic scaling-limit argument, not tested). That volatility has no long memory in any sense (the paper shows only that the autocovariance is not a power law over the observed horizon).

NOT ESTABLISHED

That H is a constant of nature; that α is zero rather than small and positive; that the roughness exists at intraday scales finer than the proxies resolve; that RFSV is preferable to a multi-scale diffusion for derivative pricing; that the finding is robust to market-microstructure noise in the proxy. Prediction is not mechanism: the forecast improvement does not validate the Hawkes explanation.

7.2 Strengths

STRENGTHS RECOGNISED BY THE AUTHORS

- Parsimony: the entire forecasting machinery uses one parameter, H , whereas AR and HAR coefficients depend on the horizon and must be re-estimated for each.
- Coherence: the same model addresses the smoothness of the path, the term structure of the ATM skew (through Fukasawa 2011), and the spurious long-memory puzzle.
- Time-homogeneity: the model is stationary, a property the authors argue is required by the stability of the shape of the volatility surface.

STRENGTHS IDENTIFIED INDEPENDENTLY

- The scaling analysis is model-free before the model is introduced: monofractality and Gaussianity are documented first and the model is written to fit them. This ordering makes the model falsifiable by Figures 3 to 11.
- Both proofs are short and verifiable, and the Appendix C algebra is correct: the Table 5 estimates were reproduced to three decimals in this review.
- The paper tests its competitor honestly in the sense that Figure 13 uses a published parameter value ($H = 0.53$) from other authors rather than a tuned straw man.
- The paper made a concrete, testable prediction about options (skew exponent $H - 1/2$) that subsequent work could and did take up (Bayer, Friz and Gatheral 2016; El Euch and Rosenbaum 2019). The fact that a one-parameter volatility model can speak to both the time series and the option surface is the reason the paper has had large influence in derivatives research.

7.3 Weaknesses

For every weakness the format is problem, why it matters, potential bias, and how it could be tested.

SINGLE MOST IMPORTANT WEAKNESS: THE PROXY AND THE ESTIMAND ARE NOT SEPARATED

Problem. The scaling regression estimates the smoothness of the proxy series. Daily realized variance equals the integrated variance plus an estimation error whose variance is not negligible, and the one-hour estimator for futures is an estimate over a short window. The paper simulates smoothing, which biases H up, but treats estimation error as absent, apart from the discretisation embedded in the uncertainty-zones model. **Why it matters.** Additive error independent of the latent process adds a constant to $m(2, \Delta)$ and flattens the log-log slope, which lowers the apparent H . **Potential bias.** Downward in H , possibly large; the two biases (smoothing up, noise down) have opposite signs and the paper quantifies only the first. **How to test.** Estimate H with noise-robust spot-volatility estimators, estimate the noise level from the signature plot or from realized-kernel versus realized-variance gaps, and fit H by quasi-likelihood on the latent process. This was done by Fukasawa, Takabatake and Westphal (2019), who conclude that after accounting for estimation

error volatility is still rough, and by Cont and Das (2024), who conclude that the apparent roughness of realized volatility originates in microstructure noise. The literature is therefore divided; section 8 gives an illustrative calculation of how much noise would be needed.

Alternative explanation: multi-scale diffusion. *Problem.* A superposition of a few Ornstein-Uhlenbeck factors with spread-out mean-reversion speeds has a covariance that, over about 1.5 decades of lag (1 to 30 days), is hard to distinguish from a function of Δ^{2H} with small H . The HAR model is itself such a superposition. *Why it matters.* The scaling evidence covers a limited log-range and the paper does not compare the likelihood of RFSV against a two- or three-factor diffusion. *Potential bias.* Overstating the evidence for fractional structure; the scaling exponent could be a summary of several time scales. *Test.* Compare out-of-sample and in-sample fit with a three-factor OU on the same proxy, and extend the lag range down to intraday scales where the one-hour futures proxy allows more resolution.

Time variation of H . *Problem.* Table 4 of the paper shows H higher in the second half for 18 of 21 indices (mean 0.113 against 0.146). *Why it matters.* The RFSV model has constant H ; if H shifts with regimes the single-parameter forecast and its calibration are mis-specified. *Potential bias.* Forecasts trained on full-sample H are too rough in calm periods and too smooth in crises. *Test.* Rolling-window estimates of H with confidence bands; structural-break tests; conditioning on the VIX regime.

Approximating α by zero. *Problem.* Proposition 3.1 is asymptotic; the simulation uses $1/\alpha = 2,000$ days against 2,000 to 3,500 sampled days. *Why it matters.* The long-horizon behaviour of the model (variance swap term structure beyond a year, stationary variance 0.34) depends on α , which is not identified from the sample. *Potential bias.* The stationary variance $\text{Var}[X] \propto \alpha^{-2H}$ is highly sensitive to α . With $H = 0.14$ a tenfold increase in α lowers $\text{Var}[X]$ by a factor $10^{0.28} = 1.9$ (own calculation). *Test.* Estimate α jointly with H by Whittle or quasi-likelihood on long samples; compare to the 2000-to-2014 sample with 2015 to 2026 data.

Forecast evaluation. *Problem.* MSE on log-variance, no equal-predictive-ability test, possible look-ahead in H and v , no loss function robust to proxy noise. *Why it matters.* Patton (2011) shows that only a family of loss functions (including QLIKE and MSE on variance) ranks forecasts correctly when the target is a noisy proxy. *Potential bias.* Overstated gains. *Test.* Diebold and Mariano (1995) tests, the model confidence set of Hansen, Lunde and Nason (2011), real-time estimation of H and v .

Gaussianity and price jumps. *Problem.* Gaussian increments are judged from histograms. Realized variance includes the contribution of jumps, which can inflate the roughness of the proxy. *Why it matters.* The lognormal formulas (10) and (13) require Gaussianity. *Test.* Quantile-quantile plots and formal tests; repeat with a jump-robust measure such as bipower variation.

Microstructure argument is heuristic. *Problem.* Section 6 chains three assumptions without a derivation and without a test in the cross-section. *Why it matters.* The explanation is presented alongside results but has far less support. *Test.* See research idea 2 in section 11.

No uncertainty quantification. No standard error is given for any estimate of H , v or P , no confidence intervals, and no p-values. A simple parametric bootstrap from the RFSV model, which the authors can already simulate, would supply them. The replication in section 5.4 shows that the sampling spread of a statistic over 3,500-day paths is large enough (0.7 to 0.8 wide for the slope, 5th to 95th percentile) that comparisons should be made with bands rather than point values.

8 Contradictions and Edge Cases

8.1 Internal inconsistencies

- The S&P smoothness is 0.142 in the text (section 2.3) and 0.124 to 0.128 in Table 3, with no explanation of the fitting range. The text range for the smoothness parameter, 0.08 to 0.2, does not match Table 3 (0.071 to 0.158) either, though it matches the half-sample table (0.059 to 0.206) only approximately.
- The weights in the forecast kernel are normalised, and the quantity $\varepsilon = 0.35$ in the paper is the normalised tail mass, but the integral defining r in the text lacks the normalising constant and equals 1.18 at $r = 1$.
- The simulation is described as consistent with the data on grounds of visual similarity (Figure 15) and the slope of $V(t)$; no dispersion is reported. The replication in section 5.4 shows that a simulated slope of 1.86 is inside, but near the upper end of, the 5th to 95th percentile range (1.70 to 1.88), so the data slope would be an unusually high draw under the stated parameters, though not an inconsistent one.

8.2 When should the result disappear?

CRITIQUE Measurement noise. If the true volatility is a Brownian-driven diffusion, $H = 0.5$, and the proxy contains additive independent noise of standard deviation s in log-volatility, then $m(2, \Delta) = v^2 \Delta^{2H} + 2s^2$ and a regression of $\log m$ on $\log \Delta$ across lags 1 to 30 returns a slope below $2H$. The table gives the apparent H for the paper's value $v = 0.3$ and various noise levels. The numbers are an illustrative calculation of this review, with an assumed noise model and with v fixed, and no sourced measure of s for the paper's proxies is available to the review.

Table 6. Apparent H from the log-log regression when the proxy contains additive noise (illustrative calculation)

Standard deviation of noise s in log-volatility	0	0.1	0.2	0.3	0.4	0.5
Apparent H if the true H is 0.50, v	0.500	0.480	0.432	0.377	0.325	0.279

Source: own calculation. Regression of $\log(v^2 \Delta^{2H} + 2s^2)$ on $\log \Delta$ for Δ the model of the noise is assumed (additive, independent, Gaussian), and generated, not estimated from data.

The reading is that noise lowers the apparent H gradually: noise of 0.5 in log-volatility brings H from 0.50 to 0.28, and reaching the observed 0.14 from a Brownian truth requires s of about 0.98, and from a truth of $H = 0.30$ requires about 0.41 (own calculation). Whether noise of that order is present in daily realized variance is an empirical matter that this review cannot settle from the

paper; the direction of the effect, however, is unambiguous, and it explains why roughness estimates from noisy proxies cannot be taken at face value without a noise correction, and why the Cont and Das (2024) argument is a serious one. The effect of smoothing in the other direction (Figure 1) partly offsets it, so the net bias on the paper's published H is of unknown sign.

CRITIQUE Extreme values of H . As H tends to $1/2$ the predictor (12) collapses to the last observation and the model becomes a Brownian log-volatility with a stationary correction, so RFSV nests the standard lognormal stochastic volatility model; as H tends to zero the one-day and one-year increments have the same scale, and the term Δ^{2H} in (13) becomes constant, so forecast uncertainty is almost independent of the horizon. For the estimated value $H = 0.14$ the model is already in this regime: the conditional standard deviation of log-variance rises by only a factor of 1.5 between one and twenty days. The same formulas give a consistency check. With the simulation parameters ($v = 0.3$, $H = 0.14$, $\alpha = 5 \times 10^{-4}$) the unconditional standard deviation of log-variance is $2 \times 0.58 = 1.17$, and the conditional standard deviation from (13) is 0.50, 0.63, 0.76 and 1.08 at horizons of 1, 5, 20 and 250 days. The model-implied ratio P is the squared ratio of these, 0.18, 0.29 and 0.42 at 1, 5 and 20 days, against the average empirical RFSV values of 0.27, 0.39 and 0.56 in Table 3 (own calculation; the parameters are the paper's simulation parameters, not index-specific estimates). The empirical error exceeds the model-implied error by 0.09 to 0.14, which is what additional proxy noise would produce, and is therefore indirect support for the noise concern above, though parameter mismatch could explain it as well. At one year the conditional dispersion of 1.08 is already close to the unconditional 1.17, so the model forgets its past within a year, and this is the point at which the parameter α , which the paper sets to zero in practice, must discipline the model

CRITIQUE Level dependence and regimes. The model has constant v and H . If volatility of volatility rises with the level of volatility (as in Heston-type models), the increments of log-volatility are not stationary in level, and the regression across lags mixes calm and turbulent episodes. The sub-sample evidence (H from 0.113 to 0.146) is the first sign of this.

CRITIQUE Other asset classes and frequencies. Everything is estimated on daily data from equity indices and two European futures. Outside this setting, the result might fail: for assets with very heavy jumps (cryptocurrencies, single stocks around earnings), the realized-variance increments are dominated by jump contributions, and H would then reflect the jump process rather than a continuous volatility path.

9 Further Literature

The table places the paper in the literature; each row gives the other paper, the shared idea, the relationship, and why it matters. Only works that the paper cites, or whose abstract was consulted for this review, are described in terms of their content.

Table 7. Literature connections

Paper	Shared idea	Relationship	Why it matters
Mandelbrot and Van Ness (1968)	Fractional Brownian motion and its moving-average representation	Preceded; theoretical foundation. The paper's equation (7.2) uses their representation for the future option-pricing work	Supplies the process whose Hurst exponent the paper estimates
Black and Scholes (1973); Dupire (1994); Heston (1993); Hull and White (1993); Hagan et al. (2002)	Constant, local and Brownian-driven stochastic volatility	Benchmarks that the paper argues are too smooth ($H = 1/2$ or constant)	The paper's claim is a rejection of the Brownian regularity assumed in these models
Ding, Granger and Engle (1993); Andersen and Bollerslev (1997); Andersen et al. (2001, 2003)	Long memory of volatility; realized volatility as a measurement device	Established the stylized fact that the paper challenges, and provided the realized-variance methodology it uses	The spurious long-memory argument is directed at the procedures in these papers
Comte and Renault (1998)	Fractional stochastic volatility with H above $1/2$	Direct predecessor; the RFSV model is the same equation with H below $1/2$ and tiny α	Shows that the novelty is a reinterpretation of parameter regions, not new dynamics
Corsi (2009)	HAR: a cascade of daily, weekly and monthly averages as an approximate long-memory predictor	Forecast benchmark that RFSV is compared to	The multi-scale alternative to a fractional kernel
Fukasawa (2011)	Skew of order $\tau^{H-1/2}$ for fractional volatility	Provides the link between the smoothness estimated here and the option surface	Turns a time-series exponent into an option-market prediction
Bacry and Muzy (2003); Duchon, Robert and Vargas (2012)	Multifractal random walk volatility and its forecast	Competing scale-invariant explanation. RFSV forecast has the multifractal predictor as the $H = 0$ limit	Both explain scaling, but multifractality predicts concave ζ_q
Jaisson and Rosenbaum (2015, 2016); Hardiman, Bercot and Bouchaud (2013)	Near-critical Hawkes processes; rough scaling limits	Supply the microstructure foundation of section 6	The mechanism is proved in these works; this paper only cites it
Bayer, Friz and Gatheral (2016); El Euch and Rosenbaum (2019); El Euch, Fukasawa and Rosenbaum (2018)	Option pricing and characteristic functions in rough models; microstructural foundations	Later, applied and extended the paper (pricing under rough volatility, rough Heston, leverage effect)	Show that the roughness concept is workable in derivatives

Paper	Shared idea	Relationship	Why it matters
Bennedsen, Lunde and Pakkanen (2022)	Brownian semistationary volatility with roughness and persistence	Qualifies the paper: finds rough and persistent volatility, with forecasting gains	Contradicts the exclusivity of the rough-without-memory interpretation
Fukasawa, Takabatake and Westphal (2019)	Quasi-likelihood estimation correcting for estimation error	Improved identification: finds volatility is rougher than previously reported	Strongest answer to the noise objection, in the paper's favour
Cont and Das (2024)	Normalised p-th variation estimator; microstructure noise	Contradiction: roughness of realized volatility arises from noise rather than from the volatility process	The most direct challenge to the main empirical claim

Source: review. Content of Fukasawa et al., Cont and Das, Bennedsen et al., and El Euch and Rosenbaum is taken from their arXiv abstracts; other rows describe works as cited in the paper itself.

10 Research Gaps

10.1 Paper-level gaps

Gap: separating roughness of volatility from noise in its proxy. *Why it matters:* it decides whether the headline finding is a property of markets or of estimators. *Existing limitation:* the paper simulates smoothing but not additive estimation error, and gives no standard errors. *Way forward:* noise-robust spot-volatility estimation, joint estimation of noise and H , parametric bootstrap from the model.

Gap: stability of H across regimes and dependence on market structure. *Why it matters:* a single H cannot serve forecasting if it moves with crises. *Existing limitation:* only a two-halves split is shown. *Way forward:* rolling and regime-conditional estimates, linked to observable liquidity and trading-activity variables.

Gap: option-price consistency. *Why it matters:* the economic payoff is joint modelling of the physical and pricing measures. *Existing limitation:* explicitly deferred. *Way forward:* the pricing programme pursued by Bayer, Friz and Gatheral (2016) and El Euch and Rosenbaum (2019).

10.2 Literature-level gaps

Gap: is volatility rough, long-memory, or both? *Why it matters:* the forecasting horizon and the term structure of risk premia differ. *Existing limitation:* Bennedsen, Lunde and Pakkanen (2022) find both roughness and persistence, the paper finds roughness without memory, and the tests used by both sides are sensitive to the proxy. *Way forward:* models in which short-scale and long-scale behaviour are parameterised separately and tested with noise-robust data.

Gap: a structural link between order flow and H. *Why it matters:* without it the Hawkes explanation is unfalsified. *Existing limitation:* calibrations of the Hawkes kernel are available for few assets and periods. *Way forward:* cross-sectional and event-based tests (research ideas 2 and 4).

10.3 Methodological gaps

Gap: estimation theory for H with noisy discrete observations of a latent path. *Why it matters:* the log-log regression has no sampling distribution. *Existing limitation:* the regression is heuristic; the quasi-likelihood and p-variation estimators of the later literature have properties that depend on assumptions about the noise. *Way forward:* compare estimators on common simulation designs and the same data. **Gap: data.** Longer samples (2015 onward) and tick data across asset classes, which the Oxford-Man library alone does not give, because its realized measures are for equity indices only (access to the library should be checked, since its continued availability could not be verified in this session).

11 Research Opportunities

Five projects follow from the paper and its critique. Data sources are named where the review has confirmed or can reasonably assume their existence; any estimate made here is labelled as such.

Idea 1. Measurement-error-robust roughness of volatility

RESEARCH QUESTION. How much of the estimated H below 0.15 survives when estimation error in the volatility proxy is modelled explicitly?

MOTIVATION. It tests the single most important weakness identified in section 7.

HYPOTHESIS. The noise-corrected H is higher than the naive H for noisy proxies and approximately unchanged for precise ones.

ECONOMIC MECHANISM. Additive proxy noise adds a constant to the second moment of increments and flattens the lag scaling (Table 6); smoothing does the opposite.

LITERATURE CONNECTION. Fukasawa, Takabatake and Westphal (2019); Cont and Das (2024); Hansen and Lunde (2006); Zhang, Mykland and Aït-Sahalia (2005).

DATA. Tick data for liquid futures and a set of single stocks; realized variance, realized kernel and pre-averaged estimators at several sampling frequencies.

VARIABLES. Log spot-volatility estimates, noise variance, sampling frequency, window length.

METHOD. Estimate H by the scaling regression and by a noise-augmented quasi-likelihood on the same series; vary the sampling frequency, since H from a correct method should not depend on it.

IDENTIFICATION STRATEGY. The variation in sampling frequency and window length changes noise and smoothing in a known way while the latent H is fixed, so the dependence of the estimated H on these choices identifies the biases.

EXPECTED CONTRIBUTION. A clear answer to whether roughness is a feature of markets or of estimators.

MAIN RISK. The conclusion could depend on the assumed noise model; tick data are costly.

Idea 2. Tick-size regime change as a natural experiment for roughness

RESEARCH QUESTION. Does a change of tick size or order-splitting regime change H , as the Hawkes-based explanation predicts?

MOTIVATION. It converts the heuristic section 6 into a test.

HYPOTHESIS. Securities with a larger tick, which constrain order splitting and quoting, have a different H from the control group after the change.

ECONOMIC MECHANISM. The tail of the Hawkes kernel, and hence H , is linked to the persistence of metaorder execution.

LITERATURE CONNECTION. Jaisson and Rosenbaum (2016); El Euch, Fukasawa and Rosenbaum (2018); the U.S. Securities and Exchange Commission Tick Size Pilot Program, which began in October 2016.

DATA. Daily and intraday trades and quotes for pilot and control stocks.

VARIABLES. Estimated H per stock and quarter, tick size group, spread, depth.

METHOD. Difference-in-differences on stock-level H estimates, with the noise-robust estimator of Idea 1.

IDENTIFICATION STRATEGY. Random assignment of stocks to pilot groups.

EXPECTED CONTRIBUTION. The first causal evidence on the origin of roughness.

MAIN RISK. The pilot was limited to small-cap stocks, whose volatility proxies are noisy, and may be too short to estimate H precisely.

Idea 3. Real-time forecasting horse race with proper loss functions

RESEARCH QUESTION. Does the RFSV forecast beat HAR and its extensions out of sample when H and v are estimated in real time and the comparison uses robust losses?

MOTIVATION. It addresses the weaknesses in section 4.3.

HYPOTHESIS. The gain over HAR shrinks but remains positive at monthly horizons, and disappears at the daily horizon.

ECONOMIC MECHANISM. A kernel with weights spread over several horizons helps most when the target horizon is long (Figure 5).

LITERATURE CONNECTION. Corsi (2009); Patton (2011); Hansen, Lunde and Nason (2011); Diebold and Mariano (1995); Bennedsen, Lunde and Pakkanen (2022).

DATA. Realized measures for equity indices, currencies and commodity futures from 2000 to the present.

VARIABLES. Realized variance at 5 minutes, forecast horizon, loss function.

METHOD. Rolling and expanding windows, QLIKE and MSE on variance, equal-predictive-ability tests and the model confidence set; add jump-robust targets.

IDENTIFICATION STRATEGY. Not causal; out-of-sample accuracy is the object. Real-time estimation removes look-ahead.

EXPECTED CONTRIBUTION. A dependable verdict on the forecasting claim, including 2008 and 2020 stress periods which the original sample only partially covers.

MAIN RISK. Gains may be too small to distinguish statistically.

Idea 4. Realized roughness against option-implied roughness

RESEARCH QUESTION. Is the time-series H equal to the skew-implied $H = 1/2 + \text{exponent}$, and does the gap price a risk premium?

MOTIVATION. The paper links the two exponents informally (0.14 against about 0.1) without testing the identity.

HYPOTHESIS. The two differ systematically, and the difference co-moves with the variance risk premium.

ECONOMIC MECHANISM. If the pricing measure changes the roughness through a change of measure on fractional volatility, the exponents under the physical and risk-neutral measures differ.

LITERATURE CONNECTION. Fukasawa (2011); Bayer, Friz and Gatheral (2016); El Euch and Rosenbaum (2019).

DATA. Option surfaces for equity indices (OptionMetrics or exchange data) and realized measures for the same underlyings.

VARIABLES. ATM skew exponent from short expiries, realized H , variance risk premium.

METHOD. Nonparametric ATM skew curves per date, a power-law fit by non-linear least squares, a rolling estimator of the time-series H , and a regression of the gap on the premium.

IDENTIFICATION STRATEGY. Cross-index and cross-time variation in both exponents; the noise-robust estimator of Idea 1 for the time-series side.

EXPECTED CONTRIBUTION. Empirical test of the main bridge between the time-series and derivatives uses of rough volatility.

MAIN RISK. Short-dated options carry microstructure noise; asymptotics hold only for very short expiries.

Idea 5. Rough versus multi-scale diffusion, a simulation-based model comparison

RESEARCH QUESTION. Can a finite-sample statistical procedure distinguish a rough fractional model from a three-factor Ornstein-Uhlenbeck volatility, given noisy proxies?

MOTIVATION. It addresses the observational-equivalence concern in section 7.3.

HYPOTHESIS. At the sample sizes of the paper, the two cannot be reliably distinguished from the scaling regression alone, but can be distinguished from intraday data.

ECONOMIC MECHANISM. A weighted sum of exponentials approximates a power law over a finite range.

LITERATURE CONNECTION. Corsi (2009); Comte and Renault (1998); Bennedsen, Lunde and Pakkanen (2022).

DATA. Simulated paths from both models, then the same real series as in Idea 3.

VARIABLES. Log spot variance with noise, sample length, lag range.

METHOD. Approximate Bayesian computation or simulation-based inference with summary statistics (scaling slopes, autocovariance shape, forecast errors); power and size of model-selection tests.

IDENTIFICATION STRATEGY. Known data-generating process in the simulation; discriminating power is measured rather than assumed.

EXPECTED CONTRIBUTION. A quantitative statement of when roughness evidence is conclusive.

MAIN RISK. Result may be negative (no discrimination at realistic sample sizes), which is informative but less publishable.

11.1 Ranking

Table 8. Ranking of research ideas (scores 1 to 5, review's own judgement)

Idea	Novelty	Feasibility	Data	Method	Importance	Total
1. Noise-robust roughness	3	4	3	5	5	20
3. Real-time forecasting horse race	2	5	4	4	4	19
4. Realized against implied roughness	4	3	3	4	4	18
2. Tick-size natural experiment	5	2	2	4	4	17
5. Rough versus multi-scale diffusion	4	3	4	3	3	17

Source: review's own subjective scoring.

The top three are ideas 1, 3 and 4. Idea 1 ranks first because it addresses the main weakness at a feasible cost with methods already in the literature, and every other idea depends on its answer. Idea 3 ranks second because it is the cheapest to execute and its result is decisive for the paper's practical claim. Idea 4 ranks third because it tests the one link that gives rough volatility its derivatives relevance and has not been settled by the paper. Idea 2 is the most novel and the best identified but is limited by data and sample size.

12 Learning Framework

12.1 Prerequisite concepts

Brownian motion, Gaussian processes and covariance functions; stationarity and Hölder continuity; the Ornstein-Uhlenbeck process; the lognormal distribution; realized variance and microstructure noise; the Black-Scholes implied volatility and the skew; autoregressive and long-memory time series; Hawkes point processes.

12.2 Core concepts learned

Roughness (Hurst exponent below 1/2) is not the same as long memory; the scaling of increment moments identifies H ; a stationary process with a very slow reversion looks like a non-stationary one on a finite sample; classical long-memory statistics can be misled by such a process; measurement design (averaging window, noise) changes the estimated smoothness.

12.3 Equations worth remembering

- Increment scaling (2) and the regression (5): $\log m(q, \Delta) = qH \log \Delta + \log(K_q v^q)$.
- Covariance of log-volatility (9) and of volatility (11): linear in Δ^{2H} and a stretched exponential respectively.
- The forecast kernel (12), a weighted average with weights summing to one.
- The smoothing bias function $f(\theta)$ in (15).

12.4 Methods learned

Moment-scaling regression for smoothness; fractional Ornstein-Uhlenbeck modelling with a vanishing mean-reversion parameter; simulation of fractional Gaussian paths (Cholesky of the covariance matrix); analytic computation of the bias from averaging; forecast comparison by the ratio of mean squared error to variance.

12.5 Questions to test understanding

COMPREHENSION

1. What does $H = 0.1$ mean for the path of log-volatility, and why is it called rough?
2. Why does the paper take α close to zero, and what is the role of Proposition 3.1?
3. Why does a daily realized-variance proxy bias the estimate of H upwards?

ANALYSIS

1. Derive $\text{Cov}[X_t, X_{t+\Delta}] = \text{Var}[X] - \frac{1}{2}v^2\Delta^{2H}$ from the scaling property, and use it to obtain the covariance of σ .
2. Show that the forecast kernel integrates to one using the Beta function.

3. Why does additive measurement noise lower the apparent H , and by how much for given v and s ?

RESEARCH

1. Design a test that would tell a rough model from a three-factor diffusion at daily frequency.
2. Which observable variable should H depend on if the Hawkes explanation is right, and in which direction?
3. How would the conclusions change if the volatility proxy came from options instead of returns?

12.6 Papers to read next

Bayer, Friz and Gatheral (2016) for pricing; El Euch and Rosenbaum (2019) for the rough Heston model; Cont and Das (2024) and Fukasawa, Takabatake and Westphal (2019) for the debate on estimation error; Bennedsen, Lunde and Pakkanen (2022) for a model with both roughness and persistence; Corsi (2009) for the HAR benchmark.

13 Conclusion

One-sentence thesis

Log-volatility measured from high-frequency data behaves like a fractional Brownian motion with a Hurst exponent near 0.1, which gives a one-parameter model that reproduces the smoothness, the apparent long memory and part of the predictability of volatility.

Three most important insights

1. Roughness and long memory are different properties, and the standard statistical tests for long memory cannot tell a rough, non-persistent process from a persistent one.
2. A single parameter, H , controls the lag structure of log-volatility, the weights of the optimal forecast, and, through Fukasawa (2011), the skew of short-dated options.
3. What is estimated is the smoothness of a proxy: averaging raises the estimate and noise lowers it, so any claim about latent volatility depends on how these two effects are treated.

Strongest evidence

The straight-line scaling of $\log m(q, \Delta)$ in $\log \Delta$ with slope proportional to q across 23 series, together with the match between data and simulated RFSV paths in the long-memory statistics.

Biggest weakness

The estimates of H come from noisy volatility proxies and the paper does not model estimation noise, which biases H downwards and is not offset by the smoothing correction it does model.

Most important research gap

Whether the roughness belongs to latent volatility or to the measurement, and whether volatility is rough, persistent, or both.

Best research opportunity

A noise-robust estimation of H across sampling frequencies and asset classes, followed by a comparison of realized roughness with option-implied roughness.

Three papers to read next

Bayer, Friz and Gatheral (2016); Cont and Das (2024); Bennedsen, Lunde and Pakkanen (2022).

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